

DSA CheatSheet

Intro to DSA:

- Algorithms Data Structures
- Time & Space Complexity
- Big O Notation

Mathematics:

- GCD/LCM
- Prime Numbers
- Sieve of Eratosthenes
- Modulo
- Bit Manipulation

Recursion & Backtracking:

- Fibonacci
- N-Queens
- Subsets
- Permutations
- Sudoku Solver

Greedy:

- Activity Selection
- Huffman Coding
- Job
- Scheduling
- Coin Change

Common Templates:

- Recursion
- Binary Search
- DP Table
- Backtracking
- Skeleton

Strings:

- Palindrome
- Anagram
- KMP
- Z Algorithm
- Manacher's
- String Matching

Stacks & Queues:

- Stack (Infix/Postfix)
- Min Stack
- Queue
- Deque
- Priority Queue (Heap)

Trees:

- Binary Tree
- BST
- AVL Tree
- Inorder/
- Preorder/Postorder
- Diameter
- LCA

Searching:

- Linear
- Binary Search
- Rotated Array
- Binary Search on Answer

Mini Projects:

- DSA Visualizer
- Algo Simulator
- Custom
- String Matcher

Tries:

- Insert
- Search
- Prefix Matching
- Word Suggestions

Linked Lists::

- Singly
- Doubly
- Circular
- Reverse
- Cycle
- Detection (Floyd's)
- Merge Point

Dynamic Programming:

- LCS
- LIS
- 0/1 Knapsack
- Matrix Chain
- Memoization
- Tabulation

Arrays:

- 1D/2D
- Prefix Sum
- Sliding Window
- Kadane's Algo
- Sorting (Bubble, Merge, Quick)

Hashing:

- HashMap
- HashSet
- Frequency Count
- Collision Handling

Graphs:

- BFS
- DFS
- Topo Sort
- Cycle Detection
- Dijkstra
- Kruskal
- DSU

Common Coding Mistakes Beginners Do:

Using = instead of == (Mistake)

- = is for assignment, == is for comparison. Forgetting Semicolons (;) (Mistake)
- Always check line endings in languages like Java, C, C++

Infinite Loops (Mistake)

- Double-check loop conditions and updates. Confusing Indexing (Mistake)
- Arrays start at 0 - count carefully!

Ignoring Edge Cases (Mistake)

- Test with empty input, large values, or duplicates.